Using an AI Agent for Network Administration Tasks

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1. Introduction

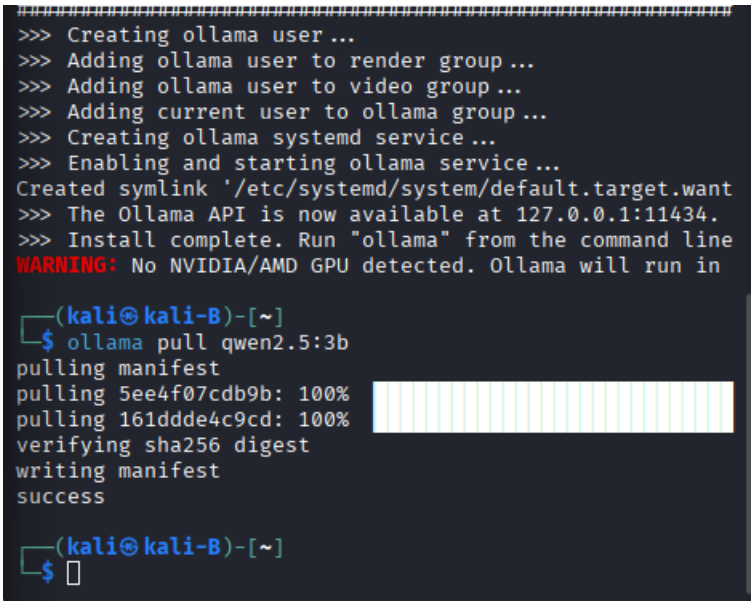
This assignment has two parts. In the first part, I practiced the Linux command line commands from Lecture 3 on my own device. In the second part, I used a small AI agent on my virtual machine and let it write and execute commands by itself. At the end, I compare the two approaches and explain why using an agent can make routine network administration tasks easier.

2. Setup

I used a Kali Linux virtual machine. On it I installed Ollama, a tool that runs AI models locally, together with a small language model called qwen2.5:3b. I wrote a short Python script that connects the model to the terminal. The way it works is:

1. I type a goal in plain English.
2. The model replies with one shell command at a time, together with a short explanation of its thinking.
3. The script shows me the command and asks for permission (Run? y/N).
4. If I approve, the script executes the command and sends the output back to the model.
5. The model reads the output and decides the next command, until it has finished the goal and prints a summary.

The model only produces text. The script is what runs the commands, and only after I approve them.



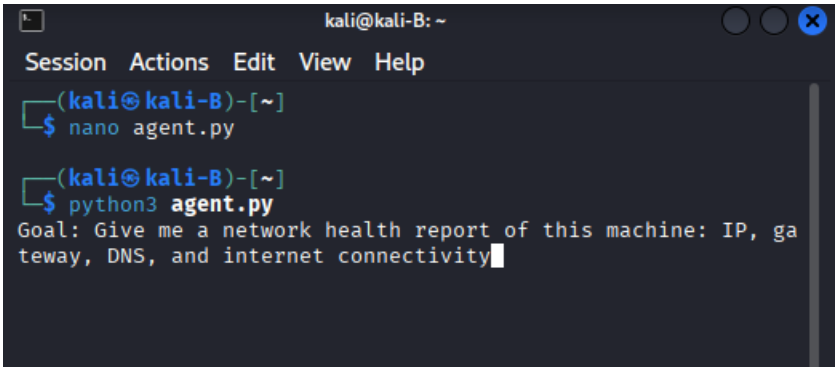
3. Part 1: Lecture 3 Commands (Manual)

I tried the Lecture 3 commands manually in a practice folder.

4. Part 2: Using an AI Agent

4.1 Task 1: Network health report

I gave the agent this goal: "Give me a network health report of this machine: IP, gateway, DNS, and internet connectivity."



Without any further instructions, the agent chose its own commands, one step at a time:

- It checked the routing table with ip route and found the default gateway (192.168.202.2) and my VM's IP address (192.168.202.129).
- It tested internet connectivity with ping -c 4 8.8.8.8, which returned 4 replies with 0% packet loss.
- It tested DNS with dig, which resolved www.google.com to an IP address.

```
┌──(kali㉿kali-B)-[~]
└─$ python3 agent.py
Goal: Give me a network health report of this machine: IP, ga
teway, DNS, and internet connectivity


[thought] First, I will check the IP address and hostname to
verify connectivity.
[command] hostname -I; hostnamectl | grep
Run? [y/N] User declined.
y

[thought] Next, I will check the gateway to see if the machin
e is reachable from outside.
[command] ip route show | grep default
Run? [y/N] default via 192.168.202.2 dev eth0 proto dhcp src
192.168.202.129 metric 100


[thought] The gateway is set to 192.168.202.2. Next, I will v
erify internet connectivity by pinging a public IP.
[command] ping -c 4 8.8.8.8
Run? [y/N] y
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=128 time=66.2 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=128 time=60.8 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=128 time=61.1 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=128 time=63.4 ms

── 8.8.8.8 ping statistics ──
4 packets transmitted, 4 received, 0% packet loss, time 3004m
s
rtt min/avg/max/mdev = 60.805/62.887/66.190/2.163 ms


[thought] The internet connectivity is confirmed as the machi
ne can reach the public IP 8.8.8.8. DNS resolution is next, s
o I will check DNS resolution by resolving a public domain na
me.
[command] dig +short www.google.com
Run? [y/N] y
216.239.38.120
```

4.2 Task 2: Lecture 3 filter commands

I then asked the agent to show me my username, hostname, current date, and the free disk space, then give me a short summary and stop.

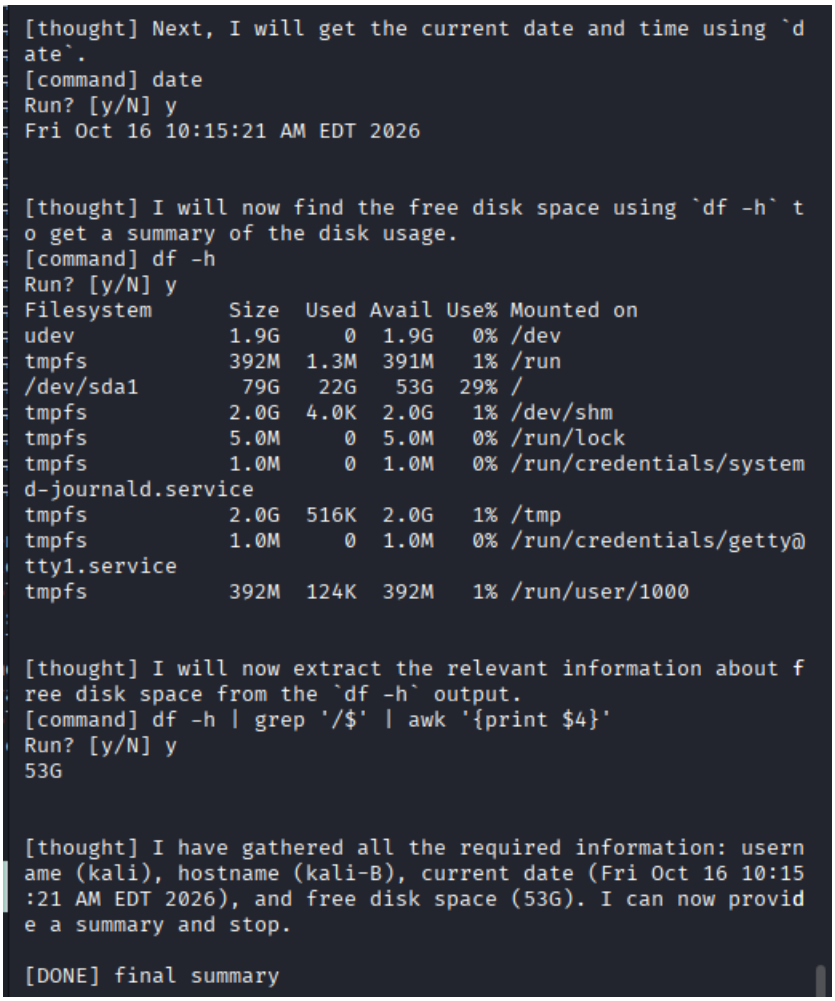
```
┌──(kali㉿kali-B)-[~]
└─$ python3 agent.py
Goal: Show me my username, hostname, current date, and the fr
ee disk space, then give me a short summary and stop.

[thought] First, I will find the hostname using `hostname`.
[command] hostname
Run? [y/N] y
kali-B


[thought] Now, I will find the username using the `whoami` co
mmand.
[command] whoami
Run? [y/N] y
kali


[thought] Next, I will get the current date and time using `d
ate`.
[command] date
Run? [y/N] y
Fri Oct 16 10:15:21 AM EDT 2026


[thought] I will now find the free disk space using `df -h` t
o get a summary of the disk usage.
[command] df -h
Run? [y/N] y
Filesystem      Size  Used Avail Use% Mounted on
udev            1.9G     0  1.9G   0% /dev
tmpfs           392M  1.3M  391M   1% /run
/dev/sda1        79G   22G   53G  29% /
tmpfs           2.0G  4.0K  2.0G   1% /dev/shm
tmpfs           5.0M     0  5.0M   0% /run/lock
tmpfs           1.0M     0  1.0M   0% /run/credentials/system
d-journald.service
tmpfs           2.0G  516K  2.0G   1% /tmp
tmpfs           1.0M     0  1.0M   0% /run/credentials/getty@
tty1.service
tmpfs           392M  124K  392M   1% /run/user/1000
```



5. Why Using the Agent Can Be Easier

- **No need to memorize syntax.** I describe what I want in plain English instead of remembering exact commands.
- **One sentence replaces many commands.**
- **It reads the output and decides the next step.**
- **It explains what it is doing.** Every command came with a short "thought", which is useful for learning why each command is used.
- **It is good for repetitive checks.** Diagnostics such as testing the gateway, internet and DNS are the same on every machine, so an agent can save time on them.

6. Problems and Limitations

The agent was helpful, but I ran into several problems, and they show that it is not the best choice for every everyday task.

- **It does not always handle errors well.** When I asked it to create a folder, the command failed because the folder already existed. The commands were chained together, so nothing after the failed one ran and the task was not completed.
- **It needs resources and setup.** Even a small 3B model needs a model download, enough RAM and disk space on the VM, and time to respond. Replies were slower than just typing a command myself, and I had to install Ollama, write the script and set up the VM first.
- **It needs clear instructions.** Long goals with many steps worked badly. Short goals that named exact folders and ended with "then stop" worked better.
- **It needs supervision.** I had to read every command before approving it, so I still needed to know the Linux commands myself.

8. Conclusion

The Lecture 3 commands are the foundation of Linux administration, and knowing them is still important for understanding and checking what an agent does. A small AI agent made multi-step tasks, such as a network health check, easier, because I could describe the goal instead of typing each command myself. However, it made mistakes, needed resources and setup, and needed constant supervision, so it is not the best choice for everyday simple tasks. It works best as an assistant for longer or repetitive diagnostics, with clear goals and human approval for every command.